



Inspiring Excellence

Brac University

Department of Electrical & Electronic Engineering

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Course Title: Engineering Project Management

Course Code: EEE359

Mid-Project

Project Title: Drone-Based Transmission Line Inspection and Vegetation Monitoring System

Group Number: 07

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Project Title:

Drone-Based Transmission Line Inspection and Vegetation Monitoring System.

Project Background/Purpose:

In Bangladesh, the supply of electricity depends on high-voltage transmission lines operated by the Power Grid Company of Bangladesh (PGCB) and the Bangladesh Rural Electrification Board (BREB). According to the official infrastructure data from PGCB the national transmission network spans a total of 18,816 circuit kilometers. This massive grid is specifically broken down into 3,087 circuit kilometers of 400 kV lines, 5,423 circuit kilometers of 230 kV lines, and 9,687 circuit kilometers of 132 kV lines. A large part of this power grid, especially the extensive 132 kV network runs through remote rural areas that are difficult to access and filled with dense trees and vegetation. Because these thousands of kilometers of lines are spread out across hard-to-reach locations, keeping electricity running reliably to rural communities is a major challenge.

Currently, inspecting these power lines depends on manual ground patrols where maintenance workers walk along the route and physically check or climb the poles. This traditional approach is extremely slow and dangerous for field staff because of fall risks and high-voltage risks. In Bangladesh, this danger is a frequent and tragic reality utility workers and linemen routinely suffer fatal electric shocks while trying to manually maintain, inspect, or clear overhead power lines (The Daily Star, 2026). Some undetected problems like cracked insulators, broken metal parts or overgrown tree branches touching the lines frequently cause sudden power outages especially during the monsoon season.

In solution this project suggests creating an automated drone-based inspection system for rural transmission line maintenance. The drone will fly along designated power line routes to capture high-resolution images while an artificial intelligence system will automatically detect damaged equipment and vegetation places. By switching from manual inspections to drone monitoring system companies can reduce the inspection costs by up to 60% . From this system we can spot dangerous faults earlier and keep human workers safe from dangerous conditions.

Objectives:

Objective Category	Simple Description	Target Outcome	Why It is Useful
1.Fly Drone instead of Working	Fly a camera drone over rural power lines to take high resolution pictures automatically.	Safely scan a 50 Km power line route.	Workers do not have to climb dangerous high voltage poles.
2. AI detects broken hardware parts	Use an AI model to automatically find broken grid parts	Catch at least 90% of broken parts correctly	Find cracked metal or broken insulators before they cause major blackouts
3. AI spots dangerous trees area	Use an AI model to spot tree branches growing too close to live electric wires.	Catches at least 80% of risky tree branches area	Detect overgrown tree places and cut branches of trees before it damaged transmission line
4. Save Time and Money	Make power line maintenance faster and reduce the cost for grid company	Cut Inspection time by 50%	Save money and reduce long power cuts and Increase efficiency of grid company
5. Smart live map monitoring System	Create an online map dashboard that pinpoints the exact GPS location of every danger place.	A working map for the repair team.	Repair teams can reach the damaged place without wasting time on searching.

Scope:

Included in the Project :

- **Rural Transmission Lines:** The project checks rural about 132 kV transmission lines and finds faults or problems in remote areas.
- **Drone Checking :** Drones are used to check the selected transmission lines.
- **Data Collection:** Collect RGB, thermal image and other needed data from the lines and nearby areas.
- **AI Fault Checking :** AI is used to find damaged or broken parts.
- **Tree and Branch Checking :** Identify trees and branches which are too close to the transmission lines.
- **GIS Mapping:** Use GIS to show faults and risky tree and branch areas on a map.
- **Safety Improvement:** Continuously checking the safety by keeping workers away from high-voltage lines.
- **Testing:** Test the system for properness , safety, and good performance.
- **Evaluation:** Analyze the results and give suggestions for future use.

Excluded from the Project :

- **Drone Making :** Drone development is not part of the project. Only use of the project ready-made or adapted drone.
- **Physical Repair :** The system will find and report the problems but will not repair them.
- **Tree Cutting :** The system will find risks with trees but will not cut them and find the other way without cutting trees.
- **New Line Building :** Building new transmission lines or towers is not covered.
- **Full Network Checking :** The project will not check the whole transmission network.
- **24/7 Monitoring :** The system will not be able to monitor the network all the time.
- **Nationwide Use :** The system will not be used across the whole country.
- **Replacing the Workers :** The system will help human workers but will not fully replace them.

Stakeholder Analysis:

Stakeholder	Types of Stakeholder	Role
Power Grid company of Bangladesh (PGCB) & Bangladesh Rural Electrification Board (BREB)	Direct	Transmission / Distribution grid operators (end owners of the inspection data and use the system)
Md. Shahriyar Mahmud Tasrif	Direct	Technical Consultant • Power & Energy Systems specialist • Validate right of way clearance • Sag standards used by the vegetation detection module
Investors	Direct	Sponsoring & Evaluates the project.
Head of Departments & CEO	Direct	Evaluator & Instructor.
Project manager	Direct	Controlling the team to bring out the maximum output from the project.
Project team	Direct	Design, build and deliver the pilot system.
Local BREB Palli Bidyut Samity (PBS)	Direct	Current manual inspectors & last users of alerts generated by the system.
Rural communities / landowners along the corridor	Indirect	Not directly involved in project work but impacted by its outcome.
Equipment suppliers	Indirect	Supplies the necessary equipment.
Staff Members	Indirect	Not directly involved in project but worked others of the company
Bangladesh Energy Regulatory Commission (BERC) & Civil Aviation Authority of Bangladesh (CAAB)	Indirect	Regulatory bodies.

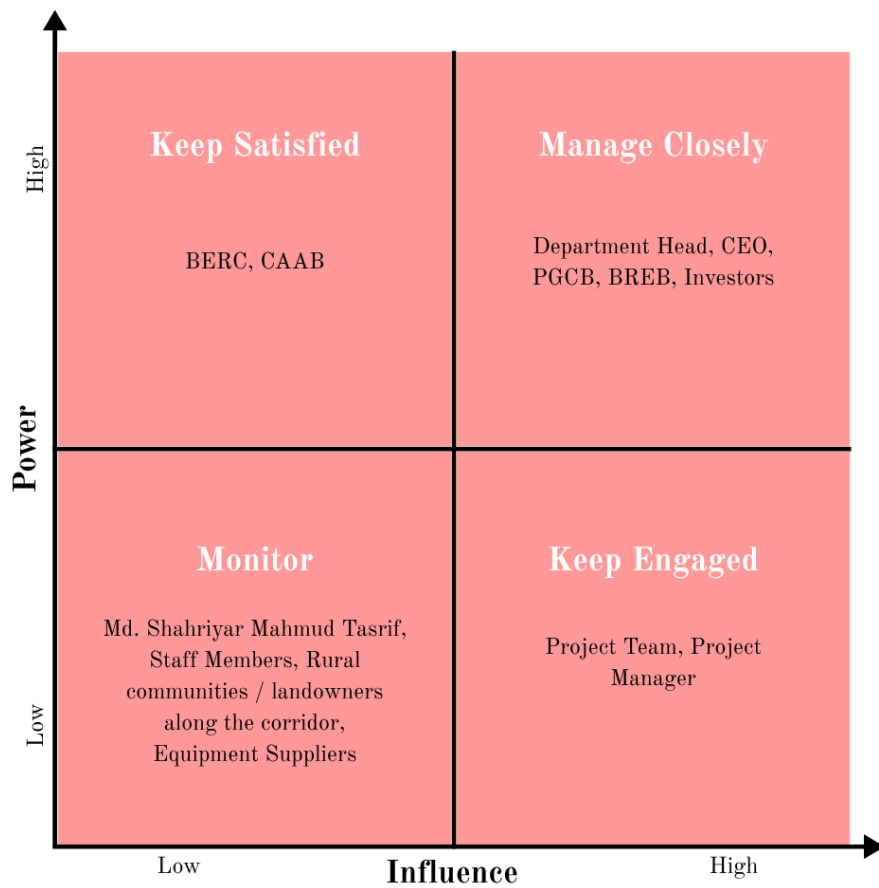


Figure :
Power-Influence Grid

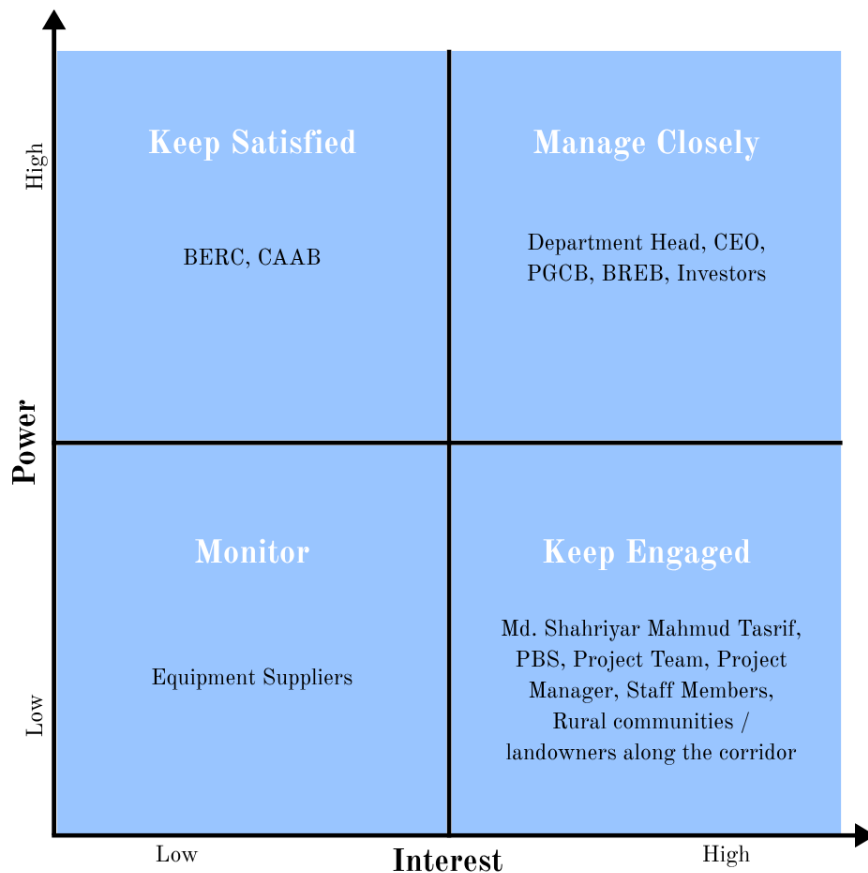


Figure :
Power-Interest Grid

Resources:

Resource Category	Description
Human	• Total 5 project team members. (Hardware, AI / software, Geographic information system or GIS, documentation, Project Manager or PM) • Technical consultant for power system and clearance standards - Md. Shahriyar Mahmud Tasrif • CAAB (Civil Aviation Authority of Bangladesh) certified one drone pilot for pilot flights
Technical / Hardware	• VTOL (Vertical Take off and landing) hybrid fixed wing UAV airframe • RGB camera gimbal • Thermal camera gimbal • RTK and GPS module • Spare batteries and ground control station • Field laptop
Software / Computing	• Flight Planning software • AI / FL framework (PyTorch / TensorFlow) • GIS platform for the dashboard (QGIS) • Cloud / local storage for image datasets • GPU enabled workstation / cloud compute credits for model training
Data / Documentation	• PGCB corridor maps • BERC / CAAB regulatory documents • Sample labelled image datasets for AI training
Financial	• Budget for hardware collections • Permits • Pilot training • contingency Details in ‘Estimated Budget’
Place of operation / Location	• Lab space • (Landowners) BREB & PBS’s permission for lands to flight access along with public corridor

Benefits and Dis-benefits:

Benefits	Dis-benefits
Faster and more frequent Transmission Line inspection coverage.	Initial high capital and Research and Development (R&D) costs.
Reduces human injury and casualties due to fall and electrocution.	Requires professional UAV pilots, cleared by Civil Aviation Authority of Bangladesh (CAAB) and other regulatory compliance.
Efficient fault detection which reduces outages and repair costs.	UAV's may get hindered due to severe disadvantageous weather conditions.
Vegetation monitoring aids in more efficient encroachment clearing.	Lesser job opportunities for manual ground maintenance staff.
Pairing with Geo-referencing (GPS tracking) ensures substantial increase in management and control.	Larger data collection may require expensive server equipment and also compromises in cybersecurity and privacy concerns.
Vegetation monitoring also includes horizontal and vertical sag clearances which reduces unforeseen outages and catastrophic failures.	Image capturing and roaming over private lands may create unwanted privacy concerns of citizens.

Key Deliverables:

SL.	Deliverables	Type
1.	UAV/Drone Prototype.	Tangible
2.	UAV Design schematics (Frame, Sensors, AI, GIS).	Intangible
3.	UAV testing data.	Intangible
4.	UAV test report.	Intangible
5.	Flight and data-capture protocol (SOP), including CAAB permit.	Intangible
6.	Vegetation encroachment detection module.	Tangible
7.	Cost-benefit, feasibility, and risk report.	Intangible
8.	Stakeholder interview report and requirements document.	Intangible
9.	Final PID / project charter and stakeholder presentation.	Intangible

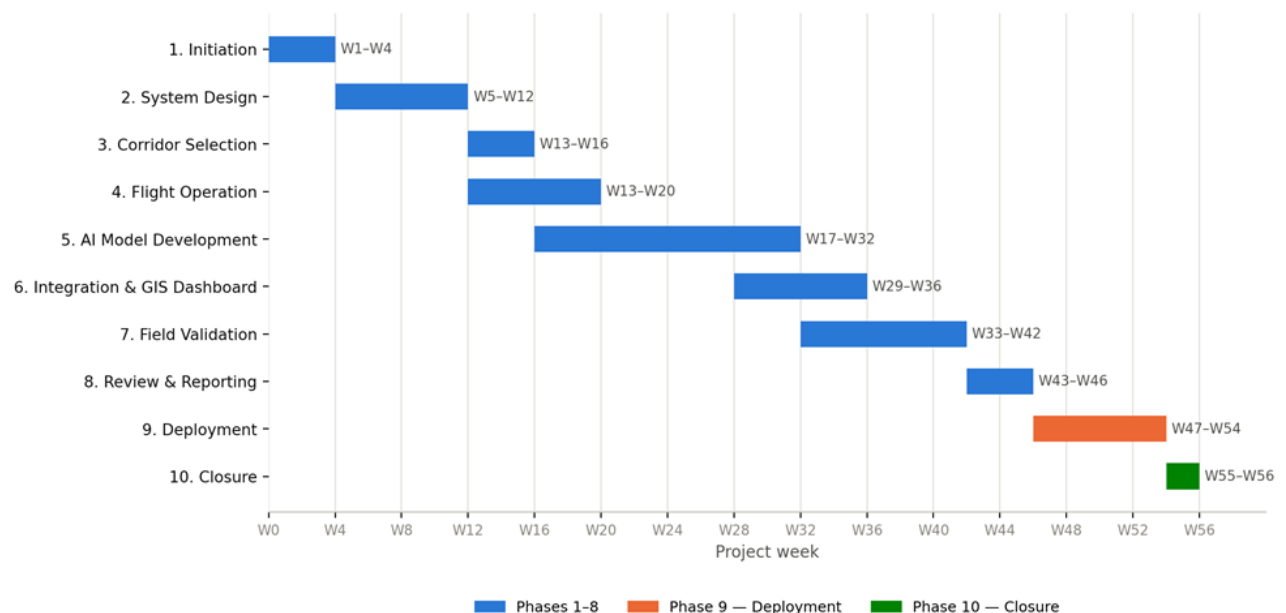
Project Team Members:

Name	Role	Responsibilities
Oshiul Alim Ornob	Project Manager	● Overall Coordination ● Regular Communicated with stakeholders ● Tracking the Schedule & Budget ● PID ownership
Alif Tamjid	UAV / Hardware Lead	● Drone platform selection ● Payload Installation & Setup ● Standard Operating Procedure (SOP) development ● Flight testing
Mahdi Zaman	AI / Software Lead	● Damage detection ● Vegetation detection model design, training & validation ● Flight testing
Souvik Barman Ratul	GIS & Data Analyst	● GIS dashboard ● Corridor Mapping ● Collecting Data, Transfer & processed the data, Fault Detection and Results Analysis
Faria Rahman Oyshi	Documentation & Risk Lead	● Report writing ● Preparing presentation ● Cost benefits analysis ● Registering the risks

Project Timeline:

Phase	Activities	Duration	Timeframe in Weeks	Dependency
1. Initiation	- Stakeholder interviews - Equipment gathering - PID finalize	4 Weeks	Week 1-4	-
2. System Design	- Sensor selection - CAAB regulatory reviews - Design in Software	8 Weeks	Week 5-12	Finish to Start with Phase 1
3. Corridor Selection	- Pilot corridor selection - Baseline survey	4 Weeks	Week 13-16	Finish to Start with phase 2
4. Flight operation	- UAV flight operations development - Image capture testing	8 Weeks	Week 13-20	Start to Start with phase 3
5. AI model Development	- AI model development - Training in damage detection, NDVI (Normalized Difference Vegetation Index) vegetation detection	16 Weeks	Week 17-32	Finish to start with phase 4 with 4 weeks lead
6. Integration & GIS dashboard	Integration & GIS dashboard development	8 Weeks	Week 29-36	Finish to Start with phase 5 with 3 weeks lead
7. Field Validation	- Pilot flights on 50 km corridor - Side by side comparison vs manual patrol	10 Weeks	Week 33-42	Finish to Start with phase 6 with 3 weeks lead
8. Review & Reporting	- Evaluation from CEO & Head of departments - Feasibility reporting - presentation	4 Weeks	Week 43-46	Finish to Start with Phase 7
9. Deployment	- Scale up planning - Pilot findings - Preparation for wider roll out	8 Weeks	Week 47-54	Finish to Start with Phase 8
10. Closure	- Hand over & sign off - Post implementation review - Benefits Realization	2 Weeks	Week 55-56	Finish to Start with Phase 9

DTLI-VMS Project Timeline — Gantt Chart (10 Phases)



Estimated Budget:

Items below have been individually compared to the current market prices for commercial use cases (hardware, software and license etc.) and an extra 10% of the subtotal has been added as contingency reserve. Cost in USD is also shown in BDT at an indicative rate of —

1 USD = 123.5 BDT (August, 2026).

Item	Estimate in USD	Estimate in BDT	Cost Type
Fixed Wing UAV(VTOL) platform (such as, YF14E-PNP) + Navigation LiDAR sensor	5,070.00	626,145	One-Time
Spare components (e.g. batteries, propellers)	1,200.00	148,200	One-Time
Camera/Sensors (dual RGB, thermal gimbal)	2,200.00	271,700	One-Time
Long-Range RC Transmitter & Telemetry Link	700.00	86,450	One-Time
Smart Battery Charger & Field Power Unit	300.00	37,050	One-Time
RTK-GPS base station & positioning kit	900.00	111,150	One-Time
Software license(s)	1,500.00	185,250	Annual
CAAB drone registration & flight permits	400.00	49,400	One-Time
Pilot training & certification	600.00	74,100	One-Time
Field logistics (e.g. travel, site access, fuel)	1,500.00	185,250	Annual
Data storage & cybersecurity	700.00	86,450	Annual
Sub-total	15,070.00	1,861,145	—
Contingency Reserve (10%)	1,507.00	186,115	One-Time
Estimated Total Cost	16,577.00	2,047,260	—

Key Risks:

Category	Risks	Mitigation Strategy
1. Regulatory	Delay in getting drone flight permit from Bangladesh Rural CAAB (Civil Aviation Authority of Bangladesh)	Contact CAAB very early in the project and submit all permit applications long before the actual flight begins.
2.Operational	Bad weather can stop our flight schedule.	Plan all the flights during the normal dry season also make extra timeslots in schedule for unexpected weather delays
3. Technical	The AI Program can make mistakes like missing broken wires or components.	A human team will carefully monitor all the pictures to detect problems and correct mistakes.
4.Technical	The drone battery dies before finishing the long flight over the rural area.	We have a VTOL hybrid fixed-wing UAV airframe system that uses less power. Also we will set multiple stop points along the route to change batteries.
5. Financial	Drone parts or software cost more money than we planned	For this reason, we will save an extra budget as an emergency budget. We will buy components after we need anything.
6. Data & Security	People might steal the drone pictures and private grid map data could leak.	We will lock all files and encrypt data by following PGCB security rules.

References:

1. Power Grid Company of Bangladesh (PGCB). *Transmission Infrastructure Information*. <https://pgcb.gov.bd/pages/static-pages/6922dd79933eb65569e152c2>
2. The Daily Star. (2026, August 12). *3 workers electrocuted to death as a crane hits a power line in Dhaka's Khilkhet*. The Daily Star. from <https://www.thedailystar.net/news/bangladesh/accidents-fires/news/3-workers-electrocuted-death-crane-hits-power-line-dhakas-khilkhet-4129506>

Project Approval Form

EEE 359

Project Title: Drone-Based Transmission Line Inspection and Vegetation Monitoring System.

Project Summary: This project develops a drone-based system to inspect transmission lines and detect damaged components and risky vegetation using AI. It aims to improve maintenance safety, reduce inspection time and costs and prevent power outages through GPS-based fault monitoring.

By signing below, the project stakeholders indicate their acceptance of the project initiation document.

Team Members:

Name	Signature	Date
Mahdi Zaman	<i>Mahdi</i>	16.08.2026
Alif Tamjid	<i>Alif</i>	16.08.2026
Souvik Barman Ratul	<i>Ratul</i>	16.08.2026
Oshiul Alim Ornob	<i>Ornob</i>	16.08.2026
Faria Rahman Oyshi	<i>Oyshi</i>	16.08.2026

Stakeholder:

Name and Designation	Signature	Date